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Spec Item	**Title**	**Description**
1	General Information	The Contractor shall provide all labor, management, supervision, tools, targeted pest control, material, and equipment required to perform Facility Investment services for facilities, ground structures, and installed equipment and systems at Range Maintenance Support, Marine Corp Base, Camp Lejeune, NC. This contract will include four hundred ninety-three (493) Training Facilities and one hundred-four (104) Live Firing Ranges.
1.1	Concept of Operations	Range maintenance tasks will consist of general repairs necessary to keep ranges and training areas in usable condition. The tasks include, but are not limited to: inspections of the Ranges, carpentry, welding, masonry, electrical, painting, and other various types of maintenance work, and transporting materials and equipment to sites. Types of work include, but are not limited to: fabricating, installing and repairing metal gates, fabricating and installing metal barriers, fabricating, installing and repairing flag poles, signs (lateral limit, range identification, impact area, no trespassing etc. metal and wood), repair/replace Range Maintenance ammo tables, wooden steps located on ranges, repair to structures/facilities, berm maintenance, firing position maintenance, bunkers, retaining walls of wood or sandbags and metal, bleachers, and wooden steps berm maintenance bunkers retaining walls, repair/replace lateral limit signs, range firing lane markers, commercial cleaning/janitorial, pressure washing, UXO/impact area grass cutting, replacing SACON wall panels, repairing rappel towers, and testing rappel points. All alterations, demolition and minor construction shall not exceed the amount of $2,000. Notification of maintenance requirements will be accomplished via written service requests (Work Authorization) for routine and urgent maintenance. The Contractor shall submit all completed Service Requests to Range Maintenance (Point of contact to be provided at time of award). The same day the work is completed or the next business day. The Contractor shall perform maintenance, repair, alteration, demolition and minor construction for the following: -Aiming Stakes (Rubber/Wood) -Firing Point (Repair/Replace) -Material Transportation (Targets, Lumber, Hulk, Jersey Barrier) -Windsocks and Frames -Ammon Tables (Metal/Wood) -Bleacher Maintenance (Metal/Wood) -Class 3 Property (Towers/Mount) -Firing Lane Markers (Repair/Replace/Dispose) -Signs -Fences -Ammo breakdown table repair -Drainage Ditches -Monuments -Flag Poles -Traffic Control Devices -Waterfront (Signs) -Inter Coastal Platforms (at entrances) -Airfields LZ's (Camp Davis, North and South)

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Spec Item	**Title**	**Description**
2	Management and Administration	
2.1	Definitions and Acronyms	Definitions and Acronyms are listed in J-1502000-01.
2.2	Personnel	The Contractor shall provide personnel with the qualifications, technical knowledge, experience and skills required for efficient operations within the Range Maintenance and Repair Support area.
2.2.1	Certification, Training, and Licensing	All maintenance and repair shall be performed by personnel trained and certified by the OEM. Personnel working with or on electrical or electronic equipment must be trained and certified per NAVFAC MO-200. Personnel conducting ground safety checks on lighting arrestors or grounding devices on facilities housing ammunition and explosives must be certified per NAVSEA OP-5. Personnel working on fire protection systems must be certified per UFC 3-600-02. Personnel performing work in HAZMAT/HAZWASTE must complete the HAZMAT/HAZWASTE handling course or have a minimum of one year of experience working with HAZMAT/HAZWASTE. All maintenance trade personnel certifying or inspecting repair or maintenance work that does not require an inspector certified by a governing directive shall be qualified at the journeyman level. Personnel Licensing Requirements: A minimum of two (2) persons with Commercial Class A CDL A minimum of one (1) person be a Licensed Welder in the State of North Carolina The Contractor shall submit proof of all certifications, training, and licensing requirements per Section F.
2.3	Special Requirements	
2.3.1	Workmanship and Material Standards	The Contractor shall be responsible for maintaining all facilities, systems, and equipment, identified in this technical sub-annex, to a standard that prevents deterioration beyond that which results from normal wear and tear and corrects deficiencies in a timely manner to assure full life expectancy of the facilities, systems, and equipment. Best commercial practices shall be applied in the performance of work. All work shall be completed per approved and accepted industry and equipment manufacturers' standards and shall comply with building and safety codes, applicable activity, local, state, and federal regulations, and other technical requirements identified within this technical sub-annex. Workmanship for maintenance and repair shall include all work necessary to complete facility and system restoration, including touch-up painting and operational checks. Upon completion of work, the Contractor shall ensure all facilities, systems, and equipment are free of missing components or defects which would affect the safety, appearance, or habitability of the facilities and structures or would prevent any electrical, mechanical, plumbing or structural system from functioning in accordance with design intent. Repairs shall be made in accordance with the manufacturers' specifications and guidelines, and standard building codes. The quality of repairs shall meet the applicable standards and shall prevent any malfunction reoccurrences caused by poor workmanship or other contractor inadequacies. The quality of the repaired areas shall be fully compatible with adjacent surfaces or equipment. Except where otherwise specified, replacements shall match existing in dimensions, finish, color, design, and functionality and shall have an appearance similar to the original finished

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Spec Item	**Title**	**Description**
		appearance with only minor unobjectionable deterioration resulting from normal use. The Contractor shall not allow debris to spread unnecessarily into adjacent areas nor accumulate in the work area. All such debris, excess material, and parts shall be cleaned up and removed at the completion of the job and at the end of each day work is in progress. Upon completion of work, any stains and other unsightly marks shall be removed.
2.3.2	Historical Preservation	Buildings and facilities designated as historical sites shall be maintained in accordance with Federal, state and local historical policies and regulations.
2.4	References and Technical Documents	References and Technical Documents are listed in J-1502000-03.

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Spec Item	**Title**	**Performance Objective**	**Related Information**	**Performance Standard**
3	Recurring Work	The Contractor shall maintain, repair, and alter facilities, and installed equipment and systems to ensure they are fully functional and in normal working condition. Re-Occurring Maintenance work for the ranges is provided in J-1502000-09. Range maintenance tasks will consist of general repairs necessary to keep ranges and training areas in usable condition. The tasks include, but are not limited to: transportation of moving equipment from one place to another, carpentry, welding, masonry, electrical, painting, wood façade repair and repainting, metal and wood prop fabrication, repair and movement, wood, metal and plastic sign fabrication, replacing and installation (Impact Area, Training Area, No Trespassing, UXO, informational etc.), wood and metal bleacher repair, inspections and other various types of maintenance work. These tasks may include the need for Architectural and Engineering Design services.	The Contractor shall provide all necessary test instruments, equipment, and tools required to perform maintenance and repair. The current list for the facilities the Contractor will be providing maintenance is provided in J-1502000-04. Site maps are provided in J-1502000-05 through J-1502000-05-18.	Facilities, ground structures, and installed equipment and systems are in normal working condition and function properly in accordance with specified standards.
3.1	Service Orders	The Contractor shall perform service order work in a timely manner and ensure	The Contractor shall receive service orders in accordance with the work reception requirements in Annex 2.	Service order work is responded to and completed within the specified time.

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Spec Item	**Title**	**Performance Objective**	**Related Information**	**Performance Standard**
		facilities, ground structures, and installed equipment and systems are restored to a safe, normal working condition and function properly.	The Contractor shall schedule and perform service orders in a way that minimize disruptions to customers and Government operations. The Contractor shall perform service orders to accomplish any work identified within the entire boundary of the installation and will include a wide variety of work. Dimensions for the Range Table Design, Lateral Limit Sign Design, and the Range Firing Lane Markers are provided in J-1502000-07 through J-1502000-07-2. Service Request Form (J-1502000-08) and the Re-occurring Range Maintenance List (J-1502000-09) are provided. Descriptions of the classifications of service orders (urgent and routine) are provided in the Definitions and Acronyms in J-1502000-01. The Contractor shall maintain sufficient materials and equipment on hand to support service order work requirements. Lack of availability of material or equipment will not relieve the Contractor from the requirement to complete service order work within the time limits specified. The Government may combine multiple repair requirements received for the same trade in the same building or structure at the same time into one service order as long as the service order threshold is not exceeded. The Contractor shall notify the KO upon identification that the	Facilities, ground structures, and installed equipment and systems are restored to normal working condition, including recertification if applicable. When repair is complete the facility, ground structure, or installed equipment and system does not present danger to personnel or equipment.

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Spec Item	**Title**	**Performance Objective**	**Related Information**	**Performance Standard**
			service order will exceed the liability limits specified below in accordance with reporting requirements in Annex 2. If Non-recurring work is issued for repairs, the Government will only pay for the portion of labor and/or material that exceeds the service order limits. The Contractor shall submit a monthly summary of completed service orders per Section F.	
3.1.1	Urgent Service Orders	The Contractor shall complete urgent service orders in a timely manner and ensure facilities, ground structures, and installed equipment and systems are restored to a safe, normal working condition and function properly. The Contractor shall respond to Urgent maintenance requests within 8 hours after receipt of a Service Request and completed within five working days. Urgent maintenance request are defined as maintenance required due to an unsafe or degraded operational training condition.	The Contractor shall perform urgent service orders to repair deficiencies without extended delay, therefore preventing further damage to facilities, ground structures, and installed equipment and systems. Performance of urgent service calls is not required outside of Government regular working hours. Refer to "Government regular working hours" in Annex 2. Urgent service orders are limited to a Recurring ceiling of 32 labor hours or $2,500 in material cost.	Urgent service orders are completed within five working days.
3.1.1.1	Range Erosion Repair	The contractor shall hydroseed eroded areas on berms, firing lines, side banks and bald spots on ranges and training areas in order to repair or prevent erosion.	The contractor shall use the Hydroseeding planting process. Using a slurry of seed and mulch applied by a truck or trailer mounted spraying device. A man-pack version can also be used for small areas.	The contractor must prepare the area to be seeded by removing rocks and other debris from its surface. Contractor shall rake the current soil or provide at least a 1-inch-deep layer of topsoil over the current soil and rake smooth to help the seeds take root. Apply seed in accordance with

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Spec Item	**Title**	**Performance Objective**	**Related Information**	**Performance Standard**
				manufacturers recommendations.
3.1.2	Routine Service Orders	The Contractor shall complete routine service orders in a timely manner and facilities, ground structures, and installed equipment and systems are restored to a safe, normal working condition and function properly. The Contractor shall respond to routine maintenance requests within 24 hours after receipt of a Service Request and complete the action within 21 days ensuring facilities and equipment, are restored to a safe, operable condition and function properly.	The Contractor shall perform routine service orders to repair deficiencies during Government regular working hours and return facilities, ground structures, and installed equipment and systems to normal working condition. Routine service orders are limited to a Recurring ceiling of 32 labor hours or $2,300 in material cost. Performance of routine service orders is not required outside of Government regular working hours. Types of work include, but are not limited to, installing and repairing metal gates, flag poles, signs (metal and wooden), erecting and maintaining barricades, fencing, ammo breakdown tables, wooden steps located on ranges, impact area maintenance and repair, repair to structures/facilities, lateral limit signs, bunkers, retaining walls of wood or sandbags and metal bleachers.	Routine service orders are completed within 21 calendar days
3.1.3	UXO Grass Cutting	Contractor shall provide UXO grass cutting quarterly in High Hazard Impact Areas aboard MCB Camp Lejeune NC	Contractor shall perform UXO Grass Cutting in High Hazard impact areas quarterly to ensure that line of sight is not impeded for occupied units on ranges across the Camp Lejeune Training Complex. Site maps are provided in J-1502000-05-15 through J-1502000-05-18.	Contractor shall cut grass no less than three (3) inches above surface grade on a quarterly basis. Prior to initiating on site activities personnel must complete 3R UXO Safety Awareness Training and a roster should be provided to the MCB Camp Lejeune Explosives Safety Officer (ESO). MCB Camp Lejeune Explosives Ordnance Disposal (EOD) on-call

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Spec Item	**Title**	**Performance Objective**	**Related Information**	**Performance Standard**
				support must be provided during UXO Grass Cutting operations.
3.1.4	Range Maintenance Quarterly Inspections	The Contractor shall complete quarterly inspections in a timely manner per Section F.	The Contractor shall perform quarterly inspections and annotate any and all deficiencies found on the ranges during Government regular working hours. The Range Locations (J-1502000-06) and Range Maintenance Inspection Checklist (J-1502000-06-1) are provided.	Inspections of the ranges are to be completed promptly.
3.1.5	Range Maintenance Annual Inspections	The Contractor shall complete annual inspections in a timely manner per Section F.	The Contractor shall perform Annual inspections and annotate any and all deficiencies found on the ranges during Government regular working hours. The Range Locations (J-1502000-06) and Range Maintenance Inspection Checklist (J-1502000-06-1) are provided.	Inspections of the ranges are to be completed promptly.
3.1.6	Range Inspection Procedures & Reporting	The contractor shall use the Range Maintenance Inspection Checklist located in Attachment J-1502000-11-2 for each Range inspected.	The contractor shall submit inspections reports and findings to the government electronically or by hard copy. The contractor shall conduct and record the maintenance inspections in enough detail so that the government will be able to create a written Service Request by interpreting the findings and communicate the repair effort needed to effect repairs. The contractor shall annotate on the checklist if the repair needs to be accomplished as soon as possible (ASAP). This would be an example of an Urgent Service Request.	Inspection Frequency: The frequency of the Range Maintenance inspections shall be Annually or Quarterly as determined by the Range Inspections Frequency List. Annual inspections to be conducted at the halfway point of the period of performance (POP). Annual inspections to be completed and findings turned into the Government within 14 days of the start date of the annual inspections.

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Spec Item	**Title**	**Performance Objective**	**Related Information**	**Performance Standard**
				Quarterly inspections to be conducted every three months starting with the beginning of the new POP for a total of (4) four inspections per POP. Quarterly inspections to be completed and findings turned into the Government within 30 days of the start date of the Quarterly inspections.
3.2	Tension Load Testing and Certifying Rappel Points	Contractor shall test and certify Rappel Towers S755, SRR211, and SRR495 and other points.	Rappel points should be tested at 5,000 lb. and in accordance with 29 CFR1910.66 Para (I)(c)(10), ANSI Z359.1, and 29 CFR1910.66 Appendix C(II)(a)(2) requirements. Camp Lejeune Rappel Towers (J-1502000-05-14) are provided.	Perform and document results. The Contractor shall submit a report of testing by a licensed certified Structural Engineer of each point tested to the Government representative assigned.

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Spec Item	**Title**	**Performance Objective**	**Related Information**	**Performance Standard**
4	Non-recurring Work	The order will specify the exact locations and types of work to be accomplished. All Non-recurring work issued will consist of the quantities that are exceeded in the Recurring work ELINs. The period of performance will be specified in each order. These tasks may include the need for Architectural and Engineering Design services.	Refer to Non-recurring ELINs for task listings, descriptions and related requirements. All periods of performance are measured from issue date of order to acceptance of the work. Performance Standards for Non-recurring work will be the same as those in Spec Item 3 where applicable. These tasks include, but are not limited to: carpentry, welding, masonry, electrical, painting, wood façade repair and repainting, metal and wood prop fabrication, repair and movement, wood, metal and plastic sign fabrication, replacing and installation (Impact Area, Training Area, No Trespassing, UXO, informational etc.), buoys, replacing SACON wall panels, wood and metal bleacher repair, inspections, commercial cleaning/janitorial, pressure washing, dust suppression and other various types of maintenance work. Work may include the Reoccurring Work provided in J-1502000-10.	Non-recurring Monthly Summary Report is submitted in a timely manner.

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SECTION F: DELIVERIES OR PERFORMANCE

DELIVERABLES						
				Distribution		
Annex/ Spec Item	Form Attachment Number	Deliverable Title	Date (s) of Submission	Original/ Copies	Number of Copies (including original)	Frequency
1502000/ 2.2.1	N/A	Certification and Licenses	Within 15 days after award or as requested by the KO	KO	1	Once for initial submittal, then as requested
3.1	J-1502000-08	Service Request Form	Daily	KO	2	Daily
3.1.4	J-1502000-10	Quarterly Range Maintenance Checklists	Quarterly	KO	2	Quarterly
3.1.5	J-1502000-10	Annual Range Maintenance Checklists	Annually	KO	2	Annually
3.2	N/A	Tension Load Testing and Certifying Rappel Points	Within 14 days after testing is completed	KO	2	As Required

ATTACHMENT J-1502000-01
DEFINITIONS AND ACRONYMS

DEFINITION	DESCRIPTION
CRANE, CATEGORY 1	Portal cranes, Hammerhead cranes, Locomotive cranes, Derricks, Floating cranes (YD), Tower cranes, Container cranes, Mobile cranes (except those indicated as category 4), including truck, cruiser, crawler, warehouse/industrial cranes, and cranes used for dragline, pile driving, clamshell, magnet, bucket work, and Aircraft crash cranes.
CRANE, CATEGORY 2 & 3	Cranes with rated capacities of 20,000 pounds or greater are category 2. Examples are Over head traveling cranes, Gantry cranes (rail mounted), Wall cranes, Jib cranes, Pillar cranes, Pillar jib cranes, Monorails and associated hoists, Fixed hoists, including chain falls. Pedestal mounted commercial boom assemblies (fixed length, telescoping, and articulating types) attached to stake trucks, trailers, flatbeds, or railcars, or stationary mounted to piers, etc., with OEM rated capacities less than 2,000 pounds.
CRANE, CATEGORY 4	Commercial truck mounted cranes, Truck mounted articulating boom cranes, Pedestal mounted commercial boom assemblies (fixed length, telescoping, and articulating types) attached to stake trucks, trailers, flatbeds, or railcars, or stationary mounted to piers, etc., with OEM rated capacities of 2,000 pounds and greater. Commercial truck mounted cranes and truck mounted articulating boom cranes with OEM capacities of 2,000 pounds and greater require a licensed operator even if the cranes are down rated below 2,000 pounds capacity for administrative purposes.
EQUIPMENT, COLLATERAL	Encompasses built-in and large substantially affixed equipment/property that is normally acquired and installed as part of a facility project.
EQUIPMENT, INSTALLED	Encompasses building-type equipment, built-in equipment, and large, substantially affixed equipment/property, and is normally acquired and installed as part of a facility project. Installed equipment is normally required to make a facility useful and operable. Removing such equipment would impair the usefulness, safety, or environment of the facility or the facility restoration work required after its removal, is substantial.
EQUIPMENT, PERSONAL PROPERTY	Personal property equipment includes all equipment other than collateral equipment. Such equipment, when acquired and used in a facility or a test apparatus, can be severed and removed aster erection or installation without substantial loss of value or damage thereto or to the premises where installed.
FACILITIES LIFE CYCLE	A facilities life cycle is divided into four stages, requirements (planning and design), acquisition (construction and acceptance), stewardship (operations, maintenance and repair), and disposal.
FACILITIES MAINTENANCE MANAGEMENT	The planning, prioritizing, organizing, controlling, reporting, evaluating, and adjusting of facilities maintenance operations to support the CNO/NAVFAC facilities policy and objectives and satisfy customers' facility needs. Defined by the International Facility Management Association as "the practice of coordinating the physical workplace with the people and work of the organization."
Integrated Maintenance Program (IMP)	IMP is a recurring state-of-the-art, reliability-centered inspection, testing, maintenance and repair program that determines best practices for managing the functions and consequences of failures of facilities equipment and system components. IMP encompasses accepted commercial practices, including reactive, preventive, predictive and proactive maintenance, into one optimal program. The IMP approach gives the Contractor full responsibility to maintain systems and equipment and perform repairs whenever necessary to ensure equipment and systems are operational and remain in a constant state of readiness. Service calls will not be issued for accomplishment of repairs on systems and equipment maintained under IMP.

LIFE-CYCLE COSTS	A form of economic analysis that considers the total cost of owning, operating, and maintaining a building or system over its useful life.
MAINTENANCE, PREVENTIVE	Maintenance designed to increase the availability of the facilities/equipment by reducing the number of unexpected breakdowns or service interruptions. It is any planned maintenance activity that improves equipment life and avoid any unplanned maintenance requirements.
MANAGEMENT INFORMATION SYSTEMS-MAINTENANCE	A computerized system that will provide sufficient information for management to evaluate differences between budgets and actual costs and evaluate performance.
REPAIR	Repair is the restoration of facilities or equipment to such a condition that it may be effectively utilized for its designated purposes by overhaul, reconstruction, or replacement of constituent parts or materials which have deteriorated by action of the elements or usage, and which have not been corrected through maintenance. This term also applies to replacement of the entire unit or system if beyond economical repair. The intent of repair is to have the equipment at normal working condition.
REPLACEMENT	Replacement, as a distinct work element, is confined to a program of planned replacement of a facility or its components. It may be further limited to major components such as air conditioning compressors, furnaces or hot water heaters. Replacement is performed when the equipment has reached the end of its useful life; when it no longer can perform due to degradation of its internal components and repair is no longer cost effective. Included under the replacement would be the major rebuilding of any component, since rebuilding also restores performance.
RESTORATION	Restoration of real property to such a condition that it can be used for its intended purpose. Includes repair or replacement work to restore facilities damaged by inadequate sustainment, excessive age, natural disaster, fire, accident or other causes.
SUSTAINMENT	Maintenance and repair activities necessary to keep a typical inventory of facilities in "normal working condition". Sustainment includes regularly scheduled maintenance as well as cyclical major repairs or replacement of components that occur periodically over the expected service life of the facilities.
SERVICE ORDER	Any work required to return a facility, system, equipment or component to normal working condition. Service orders are minor facility problem requests or requests for facilities-related work that are too small to be planned and estimated.
SERVICE ORDER CYCLE	Count down starts when the customer is notified that the work has been accepted to be accomplished to the time when the work chit is turned in by the craftsmen as complete is one complete cycle period for a service order.
SERVICE ORDER, EMERGENCY	Emergency is defined as any facility deficiency that immediately compromises the mission or life, health and safety. Always includes, but is not limited to, failure of any utility, fire protection, environmental control, or security alarm systems.
SERVICE ORDER, URGENT	Urgent is defined as any deficiency that does not immediately endanger personnel or property, but extended delays of repairs could result in damage to Government property, or soon affect the security, health, or well-being of personnel or the continued operation of a service or system.
SERVICE ORDER, ROUTINE	Routine is defined as any deficiency that does not qualify as emergency or urgent, but is needed to maintain the agreed upon facility condition. Maintain means to repair to such a condition that it may be used for its intended purpose and to normal working condition. Does not include improvements.
WEIGHT HANDLING EQUIPMENT (WHE)	Weight handling equipment consists of cranes (e.g., portal cranes, jib cranes), rigging gear (e.g., slings, shackles), and associated equipment (e.g., portable hoists, dynamometers). For purposes of this technical sub-

	annex, WHE does not include mobile or transportable truck, crawler, and railway mounted locomotive cranes covered in 1700000 BSVE.

Acronym	Title
BPVC	Boiler and Pressure Vessel Code
HVAC	Heating, Ventilation, and Air Conditioning
RPIE	Real Property Inventory Equipment
SCADA	Supervisory Control And Data Acquisition
SRM	Sustainment, Restoration and Modernization
UFC	Unified Facilities Criteria
UPV	Unfired Pressure Vessel
VTE	Vertical Transportation Equipment
UXO	Unexploded Ordnance
IMP	Information Management Program
SACON	Shock Absorbing Concrete

ATTACHMENT J-1502000-02
KNOWLEDGE, SKILL AND ABILITIES REQUIREMMENTS

A. Foreman
1. Assesses job priorities and assigns employees to appropriate job sites
2. Makes sure employees have the right tools and equipment needed for job completion
3. Oversees the work and makes sure the job runs smoothly, according to the plan and ensures the work is completed on time
4. Researches, orders and issues replacement parts, tools, material and equipment by submitting a Purchase Request (PR) to the COR
5. Helps in training new employees by guiding them as they learn new tasks
6. Gives clear directions to workers as to what needs to be done for the day's activities
7. Ensures all tools and equipment are brought back from the job site and accounted for
8. Accounts for the tools and equipment (GFE) on a monthly basis and record the inventory inspection.
9. Ensures that weekly vehicle Preventative Maintenance (PM) inspection is carried out and documented on all GFE vehicles and Heavy Equipment
10. Determines the vehicles used and routes to be taken to and from the work site
11. Holds the weekly job site safety meeting and records the meeting for future references
12. Considerable knowledge of the occupational hazards and safety precautions involved in construction and maintenance.
13. Estimates manpower requirements and materials for various projects and work assignments. Maintains inventory of materials and equipment
14. Prepares progress and other reports concerning the type of repairs made, materials used, and cost of completed work assignments
15. Considerable knowledge of a wide variety of materials, methods, practices and equipment utilized in construction and maintenance
16. Performs maintenance related work as required
17. Excellent oral and written communication skills
18. Ability to estimate manpower and material requirements; experience in preparing progress and other reports concerning the type of service(s) performed
19. Ability to establish and maintain effective working relationships with subordinates and other government and contracted personnel

B. Welder
1. Preheats and cleans metals; lays out, cuts, and fits materials for welding; selects proper welding heat and proper type and size of electrodes for correct penetration and strength of welded metals
2. Performs cutting and welding on such machinery and equipment as truck and automobile bodies, piping, pump and engine parts, trailer parts, heavy construction equipment and parts, and similar machinery and apparatus.
3. Does flat, vertical, and horizontal welding: brazes, solders, and coats nonferrous metals; inspects, brushes, and grinds welds
4. Drives truck and tows portable welder and acetylene burning-welding equipment to outside construction jobs
5. Does blacksmithing, such as manufacturing eye b9olts, braces, angle brackets, and similar items; adds metal to tools and parts
6. Maintains and makes minor repairs to welding equipment
7. Performs other maintenance related work as required
8. Considerable knowledge of the standard practices, methods, tools, and materials of the welder's trade.

\Welder (Continued)

9. Considerable knowledge of the properties of ferrous and nonferrous metals encountered in welding work
10. Ability to understand and follow oral and written instructions and to read and work from simple sketches, diagrams, and broken parts
11. Skill in the use and care of the tools and equipment common to the welding trade\

C. Carpenter

1. Repairs and maintains doors, floors, windows, partitions, ceilings, and exteriors of buildings; buildings and installs door and window framing
2. Builds wooden structures such as partitions, building, office additions,, fences, and scaffolds and does sheeting, drywall and rough carpentry on the buildings
3. Performs hand and bench work in carpentry work shop; operates planers, joiners, drill presses, power saws, sanders, and other woodworking machinery
4. Repairs and maintains hand tools, saws, and other carpentry tools; installs locks, door knobs, and other building hardware
5. Performs other maintenance related work as required.
6. Considerable knowledge of the standard practices, methods, materials, and tools of the carpentry trade
7. Considerable knowledge of the occupational hazards and safety precautions of the trade
8. Ability to plan, lay out, and perform carpentry work of average difficulty
9. Ability to work from sketches and blue prints
10. Skill in the care and use of hand and bench tools and equipment of the carpentry trade
11. Drives trucks in transporting personnel and materials to job sites

D. General Maintenance Worker

1. Inspects buildings, fixtures, machines, and other structural and/or mechanicals units, noting need for repair; performs minor maintenance, such as cleaning or lubricating
2. Prepares grades and sets forms for concrete pouring; finishes surface by hand and by machine
3. Replaces glass, screens, locks, and door latches; repairs chairs, doors, water faucets, dressers, drawers, and cabinet doors
4. Installs and repairs commodes, lavatories, and urinals
5. Makes paint, wax, and soap; loads and unloads trucks
6. Issues out and received tools; performs minor repairs on tools, machinery, and equipment; paints ladders, keeps tool room orderly and clean
7. Performs other maintenance as required
8. Considerable knowledge of the methods, practices, tools, equipment; and materials used of building and mechanical trades,
9. Some knowledge of the occupational hazards in the use of power tools and equipment and of necessary safety precautions.
10. Ability to understand and follow oral and written instructions
11. Ability to diagnose malfunctions and defects in mechanical and similar equipment and to determine adequate corrective measures
12. Ability to assign and supervise the work of a few less experienced workers
13. Skill in the use and care of hand and power tools and equipment appropriate to the area of assignment
14. Drives trucks in transporting personnel and materials to jobs

Maintenance Trades Helper

1. Assist and participates in the repair sand preventive maintenance of electric motors, switches, compressors, transformers, valves, water lines, and related equipment
2. Installs electric wiring, plumbing ducts, pipes, and fixtures; checks and tests meters,, relays, and heating and air conditioning equipment
3. Assists electricians in installing, maintaining, and repairing electrical wiring and apparatus
4. Assists auto mechanics, painters, carpenters, roofers, plumbers, plasterers, and masons by carrying materials and equipment and performing manual duties, such as mixing mortar, building scaffolds, repairing furniture, and applying asphalt tar on roofs.
5. Operates pipe-cutting and threading machines, asphalt kettles, air compressors, drills, shears, welders and other equipment.
6. Drives trucks; supplies tools, equipment, and materials to skilled tradesmen; cleans up shop
7. Digs ditches; builds forms; pours concrete
8. Trims trees, shrubs, and bushes, removes vines; operates chainsaw and uses other power equipment
9. Cuts grass using a push mower and riding mower, operates weed eater, blower and edging machines
10. Performs other maintenance related work as required
11. Some knowledge of the practices, methods, tools, materials, and equipment of the building trades
12. Knowledge of the occupational hazards of building trades and of necessary safety precautions
13. Ability to develop skill in locating and correcting defects in building installations, fixtures, and equipment.
14. Ability to understand and follow oral and written instructions
15. Skill in the use and care of tools and equipment of the building trades
16. Skill in operating trucks, fork lifts and tractors

E. Laborer

1. Loads and unloads furniture and freight from trucks; unpacks and arranges new furniture in buildings; sets up and takes down bleachers, folding chairs, and other furniture and equipment for student registrations, meetings, and other events.
2. Provides routine laboring tasks for semi-skilled and skilled workers such as setting up scaffolds, cleaning work tools, and running errands
3. Knowledge of the occupational hazards of building trades and of necessary safety precautions
4. Trims trees, shrubs, and bushes, removes vines; operates chainsaw and uses other power equipment
5. Cuts grass using a push mower and riding mower, operates weed eater, blower and edging machines
6. Cleans and sweeps sidewalks, gutters, and steps; spreads sand and cinders on icy walks and steps
7. Rakes and loads grass and brush; mows lawns and cleans grounds; waters plants and assists in the care of shrubs and grass
8. Works on a trash collection truck loading and unloading trash and wastes; washes trucks and may assist in performing maintenance tasks.
9. Digs ditches; pours concrete; repairs sidewalks. Washes walls and ceilings, moves furniture and equipment, and performs related custodial work
10. Performs other maintenance related work as required
11. Ability to understand and follow specific oral and written instructions
12. Ability to perform heavy manual labor for extended periods, often under adverse climatic conditions

Laborer (Continued)

13. Skill in operating small trucks and tractors

ATTACHMENT J-1502000-03
REFERENCES AND TECHNICAL DOCUMENTS

Reference	Title
UFC 3-430-07	Operations and Maintenance: Inspection and Certification of Boilers and Unfired Pressure Vessels
NAVFAC P-307	Management of Weight Handling Equipment
NAVFAC MO-200	Facilities Engineering Electrical Exterior Facilities
NAVSEA OP-5	Ammunition and Explosives Safety
UFC 3-600-02	O&M THE WHOLE BUILDING DESIGN GUIDE
NAVFAC MO-206	MAINTENANCE & OPERATION OF AIR COMPRESSOR PLANS
NAVFAC MO-209	MAINTENANCE OF STEAM, HOT WATER & COMPRESSED AIR
FAC MO-912	Operations, Maintenance and Repairs of Auxiliary Generators
OSHA 29 CFR 1910.140 (c)	Fall Protection Requirements
ANSI Z359	Fall Protection Code/Standards
ASTM E488	Standard Test Methods for Strength of Anchors
ASTM E575	Standard Practice for Reporting Data from Structural Tests of Building Constructions, Elements, Connections, and Assemblies
MCRP 3-01B.1	Helicopter Rope Suspension Techniques (HRST) Operations